

DEVOPS PRACTICAL – COMPLETE ANSWERS

Git, GitHub, GitLab and Docker

12 Practical Programs with Step-by-Step Commands and Explanations

Practical Topics

1. Creation of Git Repository, File Upload, Commit and Repository Management
2. Version Control using Git
3. Git Branching and Merging
4. Creating and Committing Changes
5. git init, git add, git commit and git status
6. Setting Up a Git Repository
7. GitHub Push and Pull Operations
8. GitHub Repository Cloning
9. GitLab Merge Request and Review
10. Creating a Docker Account and Setup
11. Dockerfile and Building a Web Application Image
12. Basic Docker Commands

1. Creation of Git Repository, File Upload, Commit and Repository Management

Create a local Git repository, add a file, commit the changes and push the repository to GitHub.

```
mkdir DevOpsProject
cd DevOpsProject
git init
echo "Hello DevOps" > index.txt
git status
git add index.txt
git commit -m "Added index file"
git log
```

Upload to GitHub

```
git remote add origin https://github.com/USERNAME/REPOSITORY.git
git branch -M main
git push -u origin main
```

2. Demonstrate a Practical on Version Control Tools

Git is a Distributed Version Control System (DVCS) used to track changes in source code.

Important Git commands

```
git init
git status
git add .
git commit -m "Initial commit"
git log
git branch
git remote -v
git push
git pull
```

Example

```
mkdir VCS_Demo
cd VCS_Demo
git init
echo "Version Control Demo" > demo.txt
git add .
git commit -m "First commit"
git log
```

Result: A Git repository is created and changes are tracked using commits.

3. Git Branching and Merging

```
mkdir BranchDemo
cd BranchDemo
git init
echo "Main Branch" > file.txt
git add .
git commit -m "Initial commit"
git branch feature
git checkout feature
echo "Feature Added" >> file.txt
git add .
git commit -m "Added feature"
git checkout main
git merge feature
git branch
```

Result: Changes from the feature branch are merged into the main branch.

4. Creating and Committing Changes

```
mkdir CommitDemo
cd CommitDemo
git init
echo "Hello Git" > file.txt
git status
git add file.txt
git commit -m "Added file"
echo "New changes" >> file.txt
git status
git add .
git commit -m "Updated file"
git log
```

5. Implementation of git init, git add, git commit, git status

git init

Creates a new Git repository.

```
git init
```

git add

Adds files to the staging area.

```
git add filename
git add .
```

git commit

Saves staged changes in Git history.

```
git commit -m "Initial commit"
```

git status

Shows the current repository status.

```
git status
```

Complete practical

```
mkdir GitDemo
cd GitDemo
git init
echo "DevOps Practical" > demo.txt
git status
git add .
git status
git commit -m "Added demo file"
git status
```

6. Setting Up Git Repository

Local repository

```
mkdir MyProject
cd MyProject
git init
echo "My Project" > README.md
git add README.md
git commit -m "Initial project"
```

Connect with GitHub

```
git remote add origin https://github.com/USERNAME/MyProject.git
git branch -M main
```

```
git push -u origin main
git remote -v
```

7. GitHub Operations – Push and Pull Changes

Push changes

```
git add .
git commit -m "Updated project"
git push origin main
```

Pull changes

```
git pull origin main
```

Complete example

```
git clone https://github.com/USERNAME/MyProject.git
cd MyProject
echo "New Change" >> README.md
git add .
git commit -m "Updated README"
git push origin main
```

Push: Local repository → GitHub. Pull: GitHub → Local repository.

8. GitHub Operations – Cloning a Repository

Cloning means copying a remote GitHub repository to the local computer.

```
git clone https://github.com/USERNAME/REPOSITORY.git
cd REPOSITORY
ls
git status
```

Result: The GitHub repository is successfully copied to the local system.

9. Create a Merge Request on GitLab and Review It

A Merge Request (MR) is used to request merging changes from one branch into another in GitLab.

Step 1 – Clone repository

```
git clone https://gitlab.com/USERNAME/PROJECT.git
cd PROJECT
```

Step 2 – Create branch

```
git checkout -b feature
```

Step 3 – Make changes

```
echo "New Feature" > feature.txt
```

Step 4 – Commit

```
git add .
git commit -m "Added new feature"
```

Step 5 – Push branch

```
git push -u origin feature
```

Step 6 – Create Merge Request

In GitLab: Project → Merge Requests → New Merge Request. Select Source: feature and Target: main. Add title and description, then click Create Merge Request.

Step 7 – Review

Reviewer checks code changes, commits, comments, conflicts and tests. Then Approve and Merge.

10. Create a Docker Account and Setup

Steps: Open Docker website → Sign Up → Enter email → Create username → Create password → Verify email → Login → Install Docker Desktop → Start Docker Desktop.

Verify installation

```
docker --version
docker info
```

11. Dockerfile Instructions and Build an Image for Sample Web Application

Step 1 – Create project

```
mkdir DockerWebApp
cd DockerWebApp
```

Step 2 – Create index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Docker Web App</title>
</head>
<body>
  <h1>Hello from Docker!</h1>
  <p>This is my sample web application.</p>
</body>
</html>
```

Step 3 – Create Dockerfile

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

Instruction	Purpose
FROM	Specifies the base image
COPY	Copies files into the image
EXPOSE	Specifies the application port

Step 4 – Build Docker image

```
docker build -t mywebapp .
```

Step 5 – Check image

```
docker images
```

Step 6 – Run container

```
docker run -d -p 8080:80 --name webcontainer mywebapp
```

Step 7 – Open browser

Open <http://localhost:8080>. The page should display “Hello from Docker!”

12. Basic Docker Commands

```
docker --version
docker pull nginx
```

```
docker images
docker run -d --name mycontainer nginx
docker ps
docker ps -a
docker stop mycontainer
docker start mycontainer
docker rm mycontainer
docker rmi nginx
docker logs mycontainer
docker exec -it mycontainer bash
```

Command	Purpose
docker --version	Check Docker version
docker pull	Download an image
docker images	List Docker images
docker run	Create and run a container
docker ps	List running containers
docker ps -a	List all containers
docker stop	Stop a container
docker start	Start a stopped container
docker rm	Remove a container
docker rmi	Remove an image
docker logs	View container logs
docker exec	Execute a command inside a container

QUICK REVISION – IMPORTANT COMMANDS

Git Commands

```
git init
git status
git add .
git commit -m "message"
git log
git branch
git checkout
git merge
git clone
git pull
git push
git remote -v
```

Docker Commands

```
docker --version
docker pull
docker images
docker build -t
docker run
docker ps
docker ps -a
docker stop
docker start
docker rm
docker rmi
docker logs
```

Note: Replace USERNAME, REPOSITORY and PROJECT with your actual GitHub/GitLab details before running remote commands.